

ETHAN DIGITAL ACADEMY

COMPUTER HARDWARE FUNDAMENTALS

FULL VISUAL STUDY MANUAL

Computer Hardware System

Hardware components work together as one system

Input

Keyboard, mouse, scanner

Processing

CPU and supporting circuitry

Storage

SSD/HDD and removable storage

Output

Monitor, printer, speakers

Course Code	EDA-HDW-109
Level	Beginner to Intermediate
Duration	6 Weeks (19 Lessons)
Certificate	Professional Certificate in Computer Hardware Fundamentals
Prerequisite	Computer Appreciation
Learning Format	Classroom + Safe Guided Practicals + Hardware Planning Project

Safety standard: beginner practicals focus on identification, specifications and safe external checks. Learners should not open powered equipment, handle exposed electrical parts or attempt hazardous repairs.

Course Introduction

Computer Hardware Fundamentals explains the physical components that make computers work. Learners study system types, motherboards, processors, memory, storage, input/output devices, ports, power, cooling, graphics, networking, specifications, upgrades, basic troubleshooting and preventive care.

Learning Objectives

- Identify major internal and external hardware components.
- Explain the purpose of CPU, RAM, storage, motherboard and peripherals.
- Recognise common ports and device categories.
- Read computer specifications and match them to workloads.
- Compare desktops and laptops.
- Apply safe basic troubleshooting and preventive-care principles.
- Plan a suitable computer system for a realistic use case.

Course Structure

1. Lesson 1: Introduction to Computer Hardware
2. Lesson 2: Computer System Units and Form Factors
3. Lesson 3: Motherboard Fundamentals
4. Lesson 4: Central Processing Unit (CPU)
5. Lesson 5: RAM and Working Memory
6. Lesson 6: Storage Devices: SSD, HDD and Removable Media
7. Lesson 7: Input Devices
8. Lesson 8: Output Devices
9. Lesson 9: Ports, Connectors and Cables
10. Lesson 10: Power Supply, Batteries and Electrical Protection
11. Lesson 11: Cooling, Airflow and Dust Awareness
12. Lesson 12: Graphics Hardware and Displays
13. Lesson 13: Network Hardware Basics
14. Lesson 14: Desktop and Laptop Hardware Comparison
15. Lesson 15: Reading Computer Specifications
16. Lesson 16: Hardware Compatibility and Upgrades
17. Lesson 17: Basic Hardware Troubleshooting
18. Lesson 18: Preventive Care and Hardware Records
19. Lesson 19: Final Computer Hardware Project

Lesson 1: Introduction to Computer Hardware

Introduction

Computer hardware refers to the physical parts of a computer system that can be seen or touched.

Hardware works together with software to receive input, process data, store information and produce output.

Learning Objectives

- Explain the major concepts in introduction to computer hardware.
- Complete the safe guided practical for introduction to computer hardware.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Core hardware groups include input, processing, memory, storage and output.

Key point: Internal components are housed inside the computer while peripherals connect externally.

Key point: Hardware choices should match the intended workload.

Key point: Users should not open powered equipment or attempt electrical repairs during beginner training.

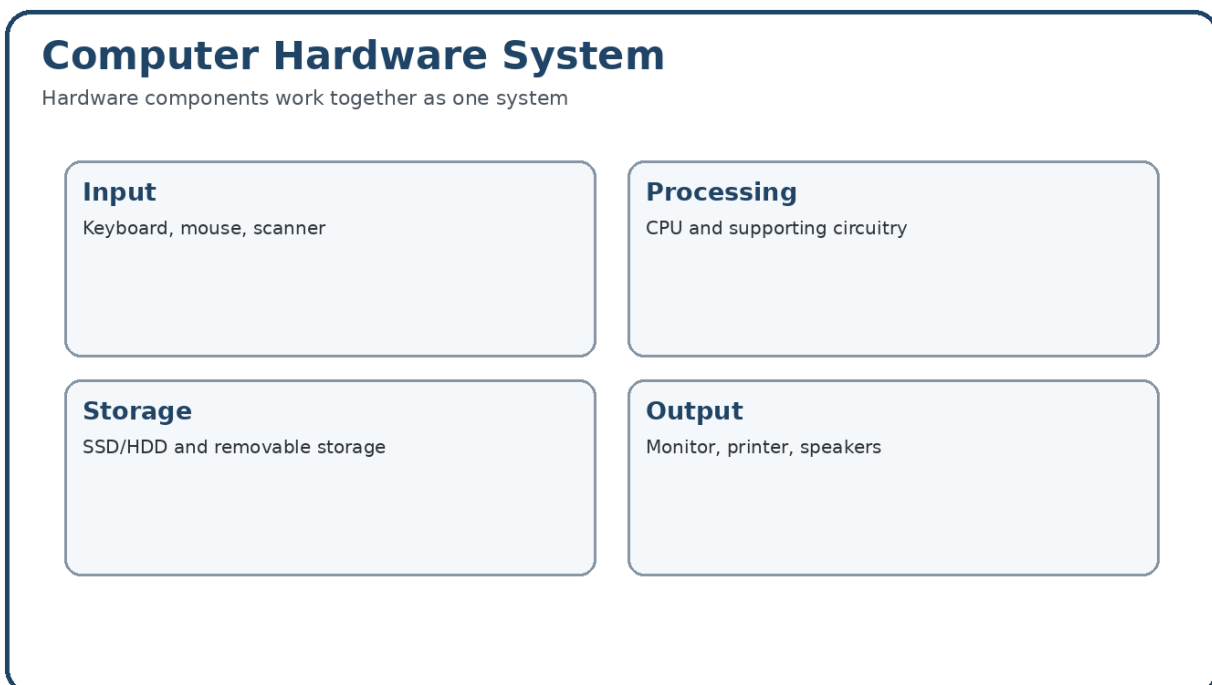


Figure 1: Instructional visual for Introduction to Computer Hardware.

Step-by-Step Guided Practical

1. Identify a desktop or laptop computer.

2. List five visible hardware components.
3. Classify each as input, output, storage or processing-related.
4. Explain one difference between hardware and software.

Ethan Digital Academy Case Study

Ethan Digital Academy can select different computer specifications for basic office training, graphics work and software-development classes.

Review Questions

20. What is the main purpose of introduction to computer hardware?
21. Name two important hardware concepts from this lesson.
22. What safety or compatibility mistake should be avoided?
23. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 1 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 2: Computer System Units and Form Factors

Introduction

Computers are built in different physical forms, including desktops, all-in-one systems, laptops, tablets and small-form-factor computers.

Learning Objectives

- Explain the major concepts in computer system units and form factors.
- Complete the safe guided practical for computer system units and form factors.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: A desktop usually separates the system unit from monitor and peripherals.

Key point: All-in-one computers integrate many components behind the display.

Key point: Laptops combine display, keyboard, pointing device, battery and internal hardware in one portable unit.

Key point: Form factor affects portability, upgrade options, cooling and available ports.

Step-by-Step Guided Practical

1. Compare a laptop and desktop in the classroom.
2. Record portability, screen, keyboard, ports and power source.
3. Identify which system is easier to move.
4. Choose a form factor for a computer lab and explain why.

Ethan Digital Academy Case Study

A fixed Academy lab may favour desktops, while instructors who travel between classrooms may need laptops.

Review Questions

24. What is the main purpose of computer system units and form factors?
25. Name two important hardware concepts from this lesson.
26. What safety or compatibility mistake should be avoided?
27. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 2 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 3: Motherboard Fundamentals

Introduction

The motherboard is the main circuit board that connects the processor, memory, storage interfaces, expansion devices and external ports.

Learning Objectives

- Explain the major concepts in motherboard fundamentals.
- Complete the safe guided practical for motherboard fundamentals.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: The CPU socket or package area holds the processor.

Key point: RAM slots accept compatible memory modules.

Key point: Expansion slots may support graphics, network or other add-in cards.

Key point: Firmware stored on the system board helps initialise hardware during startup.

Key point: Motherboards must match the processor, memory type, case and other components.

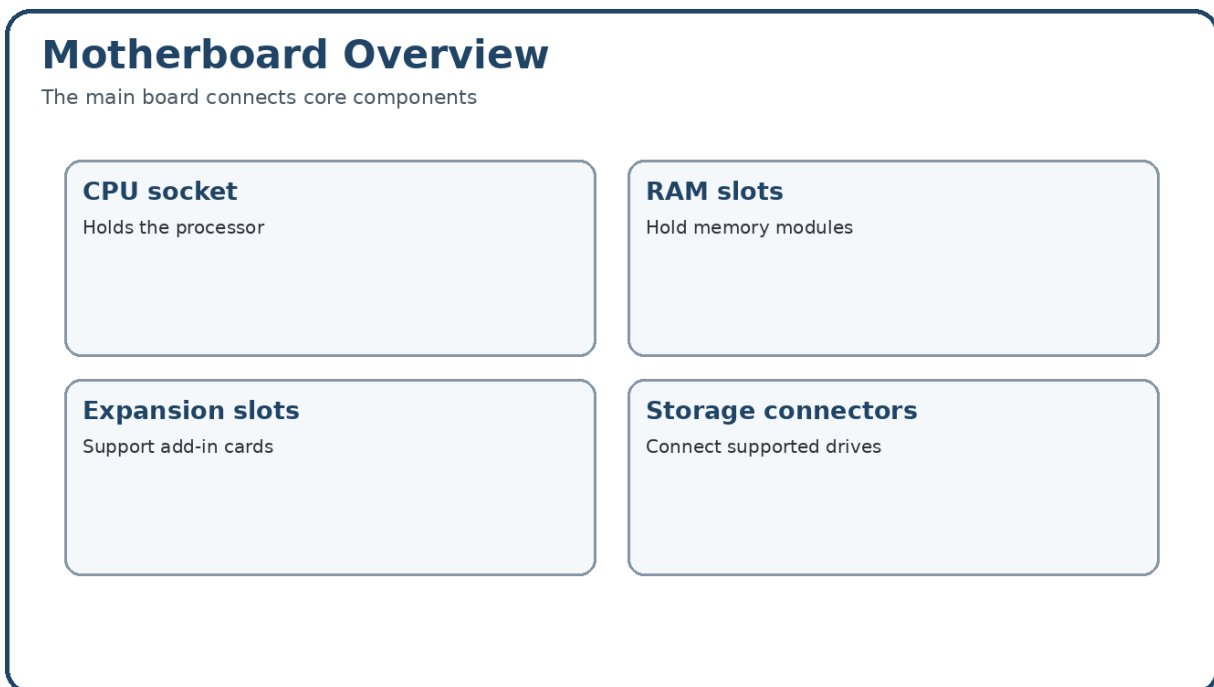


Figure 3: Instructional visual for Motherboard Fundamentals.

Step-by-Step Guided Practical

1. Use an instructor-provided motherboard diagram or powered-off display unit.
2. Locate the CPU area, RAM slots, expansion slots and storage connectors.
3. Do not remove components unless the instructor has prepared a safe training unit.
4. Label the parts on a worksheet.

Ethan Digital Academy Case Study

Understanding motherboard layout helps Academy learners read hardware specifications without treating the computer as a mysterious box.

Review Questions

28. What is the main purpose of motherboard fundamentals?
29. Name two important hardware concepts from this lesson.
30. What safety or compatibility mistake should be avoided?
31. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 3 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 4: Central Processing Unit (CPU)

Introduction

The CPU is the main general-purpose processor. It executes instructions and coordinates many operations in a computer.

Learning Objectives

- Explain the major concepts in central processing unit (cpu).
- Complete the safe guided practical for central processing unit (cpu).
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Processor performance depends on architecture, generation, core count, clock behaviour, cache and workload.

Key point: More cores can help tasks designed to use them, but core count alone does not determine overall speed.

Key point: The CPU produces heat and normally requires an appropriate cooling solution.

Key point: Processor compatibility with the motherboard is essential.

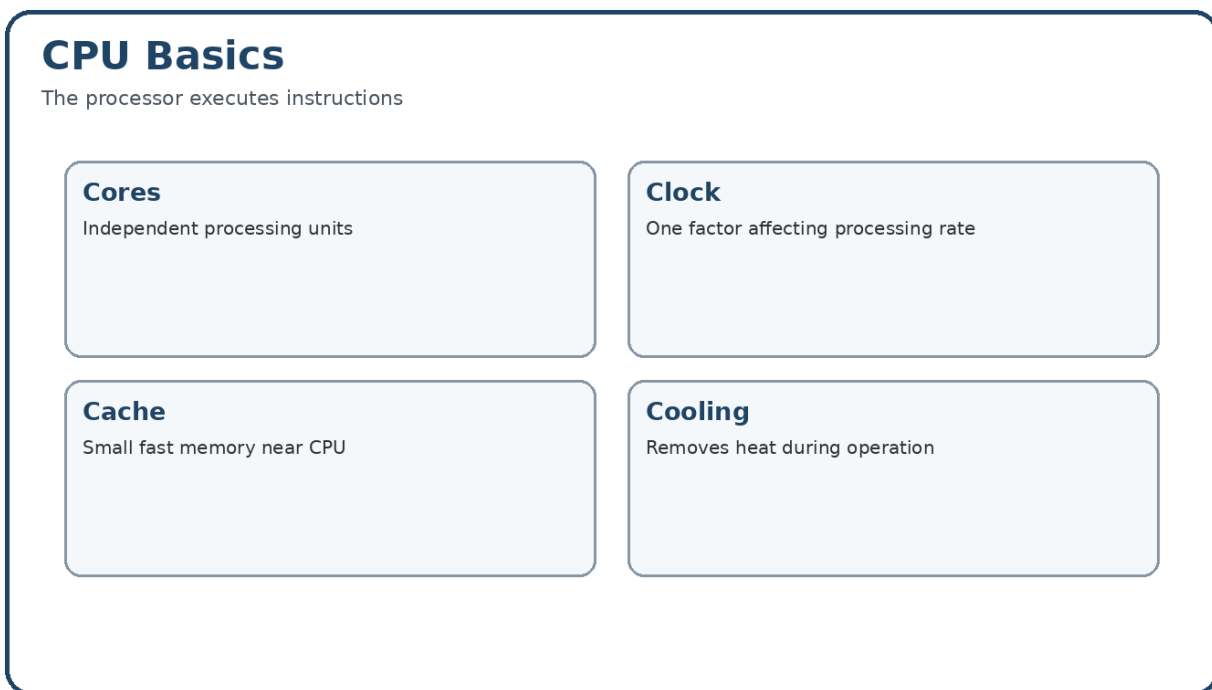


Figure 4: Instructional visual for Central Processing Unit (CPU).

Step-by-Step Guided Practical

1. Open the operating system's system-information page.
2. Record the processor name shown.
3. Identify the number of cores or logical processors if displayed.
4. Compare two fictional processor specifications supplied by the instructor.

Ethan Digital Academy Case Study

Basic office systems and demanding creative workstations may require different processor capabilities.

Review Questions

32. What is the main purpose of central processing unit (cpu)?
33. Name two important hardware concepts from this lesson.
34. What safety or compatibility mistake should be avoided?
35. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 4 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 5: RAM and Working Memory

Introduction

Random Access Memory (RAM) is fast temporary memory used for programs and data that are actively in use.

Learning Objectives

- Explain the major concepts in ram and working memory.
- Complete the safe guided practical for ram and working memory.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: RAM is volatile, so its contents are normally lost when power is removed.

Key point: More RAM can improve multitasking when a system is running short of memory.

Key point: Memory modules must be compatible with the motherboard and processor platform.

Key point: Capacity is commonly expressed in gigabytes.

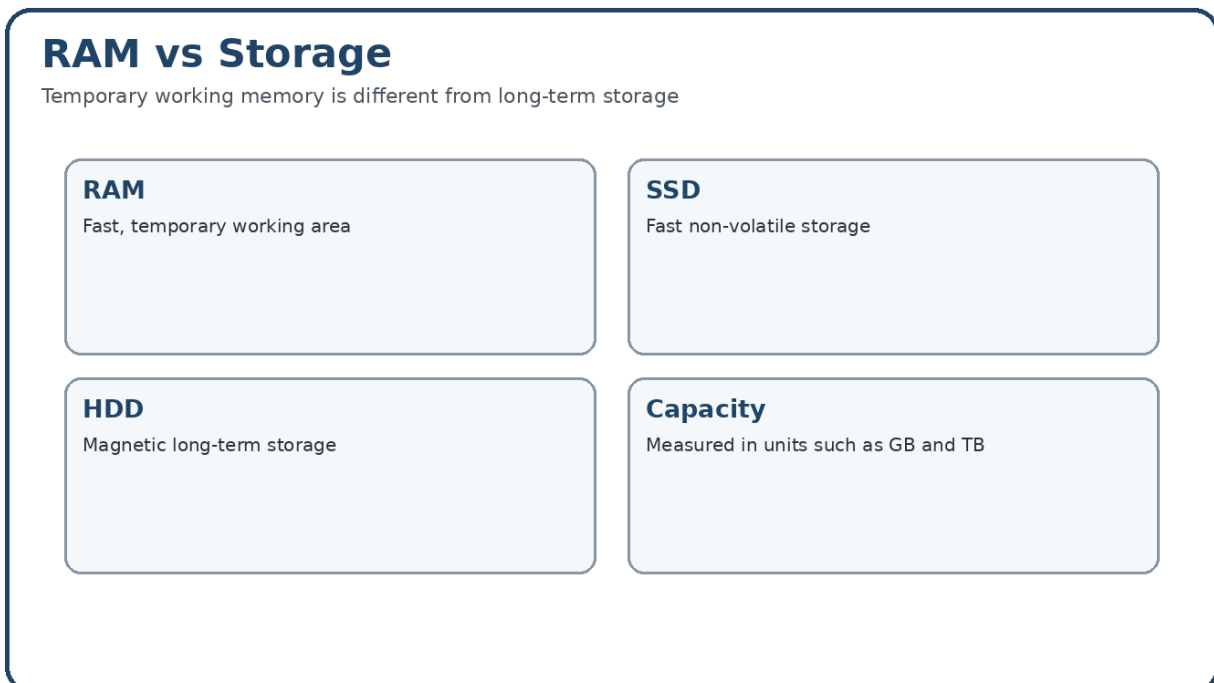


Figure 5: Instructional visual for RAM and Working Memory.

Step-by-Step Guided Practical

1. Open system information or Task Manager on an authorised computer.

2. Record installed RAM.
3. Open two normal applications and observe memory use.
4. Close one application and observe the change.

Ethan Digital Academy Case Study

A computer used for many browser tabs, office applications and video meetings may benefit from more RAM than a very basic single-task system.

Review Questions

36. What is the main purpose of ram and working memory?
37. Name two important hardware concepts from this lesson.
38. What safety or compatibility mistake should be avoided?
39. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 5 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 6: Storage Devices: SSD, HDD and Removable Media

Introduction

Storage keeps files and software when the computer is powered off. Common technologies include solid-state drives and hard disk drives.

Learning Objectives

- Explain the major concepts in storage devices: ssd, hdd and removable media.
- Complete the safe guided practical for storage devices: ssd, hdd and removable media.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: SSDs use flash memory and generally provide faster access with no moving read/write mechanism.

Key point: HDDs store data magnetically on rotating platters and can offer large capacity at lower cost.

Key point: USB flash drives and memory cards are removable storage but should not be the only copy of important work.

Key point: Storage capacity and performance are separate considerations.

Step-by-Step Guided Practical

1. Open File Explorer or the system storage page.
2. Record the main drive's total and free capacity.
3. Identify whether the system reports SSD/HDD where available.
4. Copy a harmless practice file to an authorised removable drive, then safely eject it.

Ethan Digital Academy Case Study

Academy course materials need adequate storage, while important administrative files should also follow proper backup procedures.

Review Questions

40. What is the main purpose of storage devices: ssd, hdd and removable media?
41. Name two important hardware concepts from this lesson.
42. What safety or compatibility mistake should be avoided?
43. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 6 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 7: Input Devices

Introduction

Input devices allow users or sensors to send data and commands into a computer.

Learning Objectives

- Explain the major concepts in input devices.
- Complete the safe guided practical for input devices.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Keyboards enter text and commands.

Key point: Mice and touchpads control pointers and selections.

Key point: Scanners convert physical documents into digital images.

Key point: Microphones and webcams capture audio and video.

Key point: Touchscreens combine input with visual output.

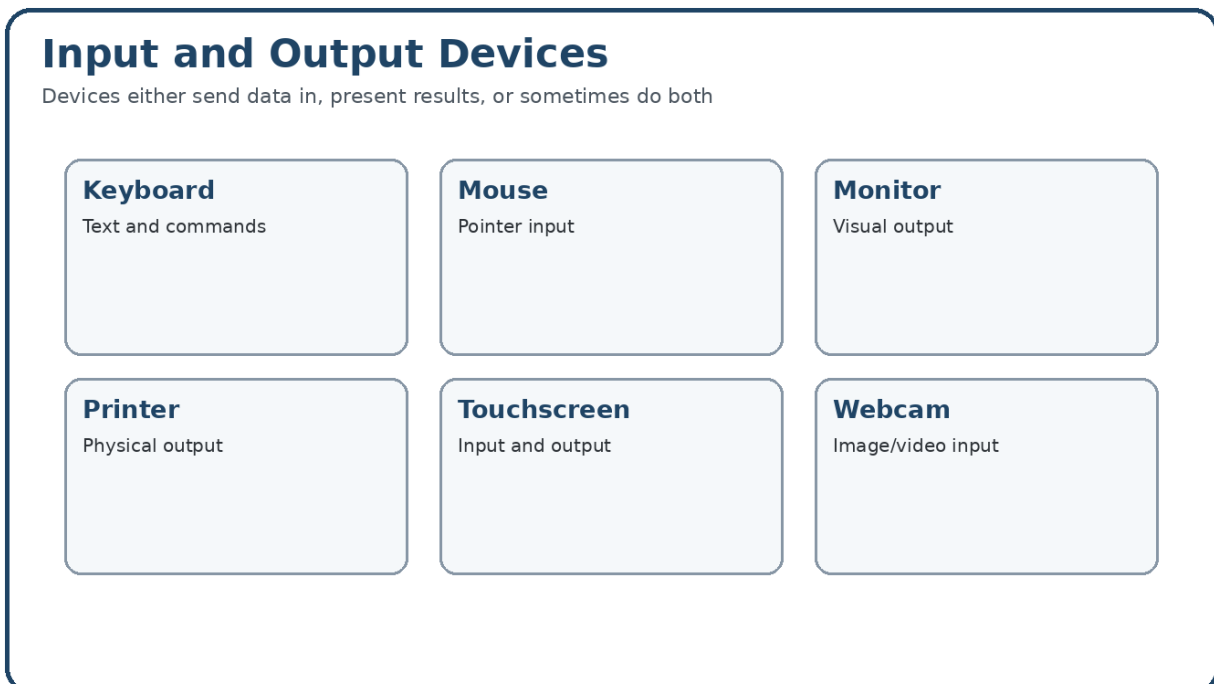


Figure 7: Instructional visual for Input Devices.

Step-by-Step Guided Practical

1. List the input devices connected to your computer.
2. Use keyboard and mouse to create a short document.
3. Test a webcam or microphone only if authorised.
4. Explain what data each device sends to the computer.

Ethan Digital Academy Case Study

Different Academy courses may use webcams for online classes, scanners for document workflows and graphics tablets for creative work.

Review Questions

44. What is the main purpose of input devices?
45. Name two important hardware concepts from this lesson.
46. What safety or compatibility mistake should be avoided?
47. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 7 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 8: Output Devices

Introduction

Output devices present processed information to users or another medium.

Learning Objectives

- Explain the major concepts in output devices.
- Complete the safe guided practical for output devices.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Monitors display visual information.

Key point: Printers produce physical copies.

Key point: Speakers and headphones provide audio output.

Key point: Projectors display a computer image to a larger audience.

Key point: Output quality depends on both device capability and software settings.

Step-by-Step Guided Practical

1. Identify the monitor's display settings.
2. Locate the volume control for audio output.
3. Inspect printer options without printing unnecessarily.
4. List one classroom use for a projector.

Ethan Digital Academy Case Study

An Academy instructor can connect a laptop to a projector or large display for demonstrations.

Review Questions

48. What is the main purpose of output devices?
49. Name two important hardware concepts from this lesson.
50. What safety or compatibility mistake should be avoided?
51. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 8 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 9: Ports, Connectors and Cables

Introduction

Ports connect computers to peripherals, displays, networks and power. Correct identification prevents damage and connection errors.

Learning Objectives

- Explain the major concepts in ports, connectors and cables.
- Complete the safe guided practical for ports, connectors and cables.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: USB is widely used for peripherals and data devices.

Key point: HDMI commonly carries digital video and audio.

Key point: Ethernet ports support wired networking.

Key point: Audio connectors may support headphones, speakers or microphones.

Key point: Connector shapes and capabilities vary; never force a plug into a port.

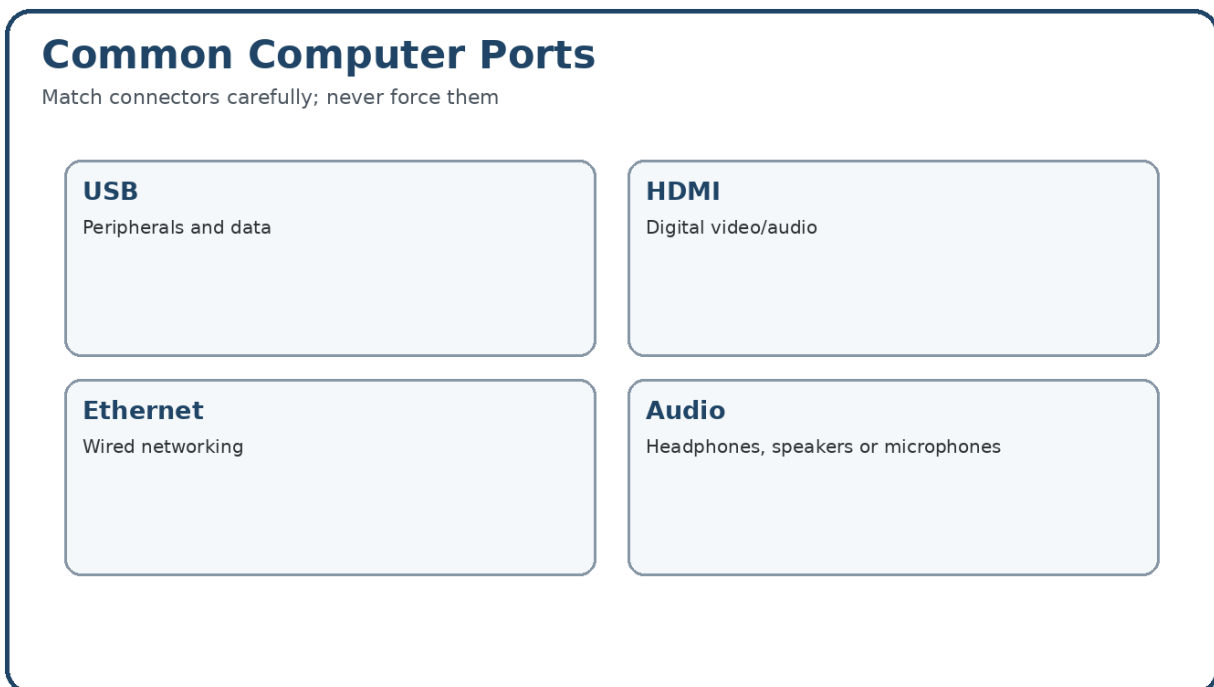


Figure 9: Instructional visual for Ports, Connectors and Cables.

Step-by-Step Guided Practical

1. Inspect the outside of an authorised computer.
2. Identify USB, display, network and audio ports that are present.
3. Match sample cables to their correct ports without forcing them.
4. Record which ports are absent on the device.

Ethan Digital Academy Case Study

Before buying a projector or external monitor, the Academy should confirm that the computer has a compatible display connection.

Review Questions

52. What is the main purpose of ports, connectors and cables?
53. Name two important hardware concepts from this lesson.
54. What safety or compatibility mistake should be avoided?
55. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 9 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 10: Power Supply, Batteries and Electrical Protection

Introduction

Computers require stable electrical power. Desktop power supplies convert incoming electricity for internal components, while laptops commonly use adapters and batteries.

Learning Objectives

- Explain the major concepts in power supply, batteries and electrical protection.
- Complete the safe guided practical for power supply, batteries and electrical protection.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: The desktop PSU distributes appropriate power to internal hardware.

Key point: Laptop batteries provide portable power and gradually lose capacity with age.

Key point: A surge protector can help protect equipment from voltage spikes.

Key point: A UPS can provide temporary backup power and time for safe shutdown.

Key point: Electrical repairs and damaged power equipment should be handled by qualified adults or technicians.

Power and Protection

Safe power practices protect people and equipment

PSU

Converts incoming power for computer components

Surge protection

Helps protect against voltage spikes

UPS

Can provide temporary backup power

Shutdown

Use normal operating-system shutdown

Figure 10: Instructional visual for Power Supply, Batteries and Electrical Protection.

Step-by-Step Guided Practical

1. Identify the computer's external power connection without opening equipment.
2. Locate battery status on a laptop if applicable.
3. Demonstrate normal operating-system shutdown.
4. Write a safe response to a damaged power cable: stop using it and report it.

Ethan Digital Academy Case Study

A computer lab should use suitable electrical protection and normal shutdown procedures, especially where power supply can be unstable.

Review Questions

56. What is the main purpose of power supply, batteries and electrical protection?
57. Name two important hardware concepts from this lesson.
58. What safety or compatibility mistake should be avoided?
59. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 10 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 11: Cooling, Airflow and Dust Awareness

Introduction

Electronic components generate heat. Cooling systems move heat away so the computer can operate within safe temperatures.

Learning Objectives

- Explain the major concepts in cooling, airflow and dust awareness.
- Complete the safe guided practical for cooling, airflow and dust awareness.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Fans move air through a case or laptop cooling path.

Key point: Heatsinks increase surface area for heat transfer.

Key point: Blocked vents can reduce cooling effectiveness.

Key point: Dust can restrict airflow over time.

Key point: Beginner learners should clean only external surfaces or filters when instructed; internal servicing should be done by authorised persons.

Cooling and Airflow

Heat management supports reliable operation

Fans

Move air through the case

Heatsink

Transfers heat away from chips

Air vents

Need clear airflow

Dust control

Cleaning should be done safely by authorised persons

Step-by-Step Guided Practical

1. Locate external air vents on a powered-off computer.
2. Check that vents are not blocked by books or fabric.
3. Listen for normal fan operation after startup.
4. Write three ways to support good airflow without opening the computer.

Ethan Digital Academy Case Study

Academy lab computers should be positioned so ventilation openings remain clear.

Review Questions

60. What is the main purpose of cooling, airflow and dust awareness?
61. Name two important hardware concepts from this lesson.
62. What safety or compatibility mistake should be avoided?
63. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 11 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 12: Graphics Hardware and Displays

Introduction

Graphics hardware produces images for the display. Some computers use graphics integrated with the processor, while others have a separate graphics processor.

Learning Objectives

- Explain the major concepts in graphics hardware and displays.
- Complete the safe guided practical for graphics hardware and displays.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Integrated graphics can be suitable for office work and general learning.

Key point: Dedicated GPUs can provide more performance for demanding graphics, video and 3D workloads.

Key point: Display resolution describes the number of pixels used to form the image.

Key point: Refresh rate describes how often the display updates each second.

Key point: Hardware should be selected according to the actual software workload.

Step-by-Step Guided Practical

1. Open display settings.
2. Record screen resolution and scaling.
3. Identify the graphics adapter from system information if available.
4. Compare a fictional office PC and creative workstation specification.

Ethan Digital Academy Case Study

Canva and office training may run well on modest graphics hardware, while advanced video or 3D courses may require stronger graphics capability.

Review Questions

64. What is the main purpose of graphics hardware and displays?
65. Name two important hardware concepts from this lesson.
66. What safety or compatibility mistake should be avoided?
67. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 12 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 13: Network Hardware Basics

Introduction

Network hardware connects computers to local networks and the internet.

Learning Objectives

- Explain the major concepts in network hardware basics.
- Complete the safe guided practical for network hardware basics.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: A network interface can be wired Ethernet or wireless Wi-Fi.

Key point: Routers direct traffic between networks and commonly provide local connectivity.

Key point: Switches connect multiple wired devices within a local network.

Key point: Modems or provider equipment connect a site to an internet service depending on the technology used.

Key point: Network devices should be configured only by authorised administrators.

Step-by-Step Guided Practical

1. Identify whether your computer is using Wi-Fi or Ethernet.
2. Locate the network icon and connection name.
3. Inspect an Ethernet cable if available.
4. Draw a simple computer-router-internet diagram.

Ethan Digital Academy Case Study

An Academy lab can use wired networking for fixed desktops and Wi-Fi for authorised mobile devices.

Review Questions

68. What is the main purpose of network hardware basics?
69. Name two important hardware concepts from this lesson.
70. What safety or compatibility mistake should be avoided?
71. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 13 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 14: Desktop and Laptop Hardware Comparison

Introduction

Desktops and laptops perform similar computing functions but package their components differently.

Learning Objectives

- Explain the major concepts in desktop and laptop hardware comparison.
- Complete the safe guided practical for desktop and laptop hardware comparison.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Desktops are often more modular and may offer easier upgrades.

Key point: Laptops prioritise portability and integrate battery, display and input devices.

Key point: Laptop components can be model-specific and less accessible.

Key point: Cooling and power limits differ between compact and larger systems.

Key point: Purchase decisions should consider mobility, repairability, ports and workload.

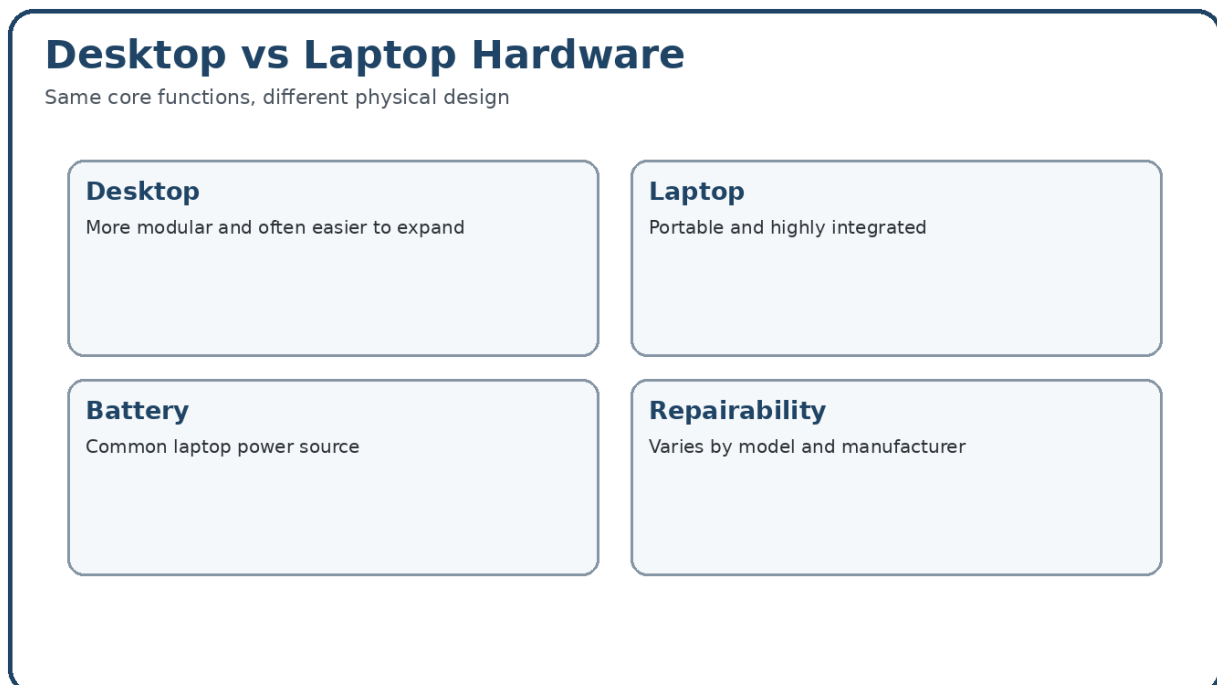


Figure 14: Instructional visual for Desktop and Laptop Hardware Comparison.

Step-by-Step Guided Practical

1. Create a comparison table with Desktop and Laptop columns.
2. Compare portability, upgradeability, battery, display and ports.
3. Choose one for a travelling instructor.
4. Choose one for a fixed computer lab and justify both decisions.

Ethan Digital Academy Case Study

Ethan Digital Academy may use desktops in a permanent lab while staff use laptops for presentations and field training.

Review Questions

72. What is the main purpose of desktop and laptop hardware comparison?
73. Name two important hardware concepts from this lesson.
74. What safety or compatibility mistake should be avoided?
75. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 14 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 15: Reading Computer Specifications

Introduction

A specification sheet describes the main hardware capabilities of a computer. Buyers should interpret the whole system rather than focus on one impressive number.

Learning Objectives

- Explain the major concepts in reading computer specifications.
- Complete the safe guided practical for reading computer specifications.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Processor model and generation help identify capability.

Key point: RAM capacity affects multitasking.

Key point: Storage type and capacity affect responsiveness and file space.

Key point: Display size/resolution, graphics hardware, ports and battery life may matter depending on use.

Key point: Warranty, support and upgrade options are also practical purchase factors.

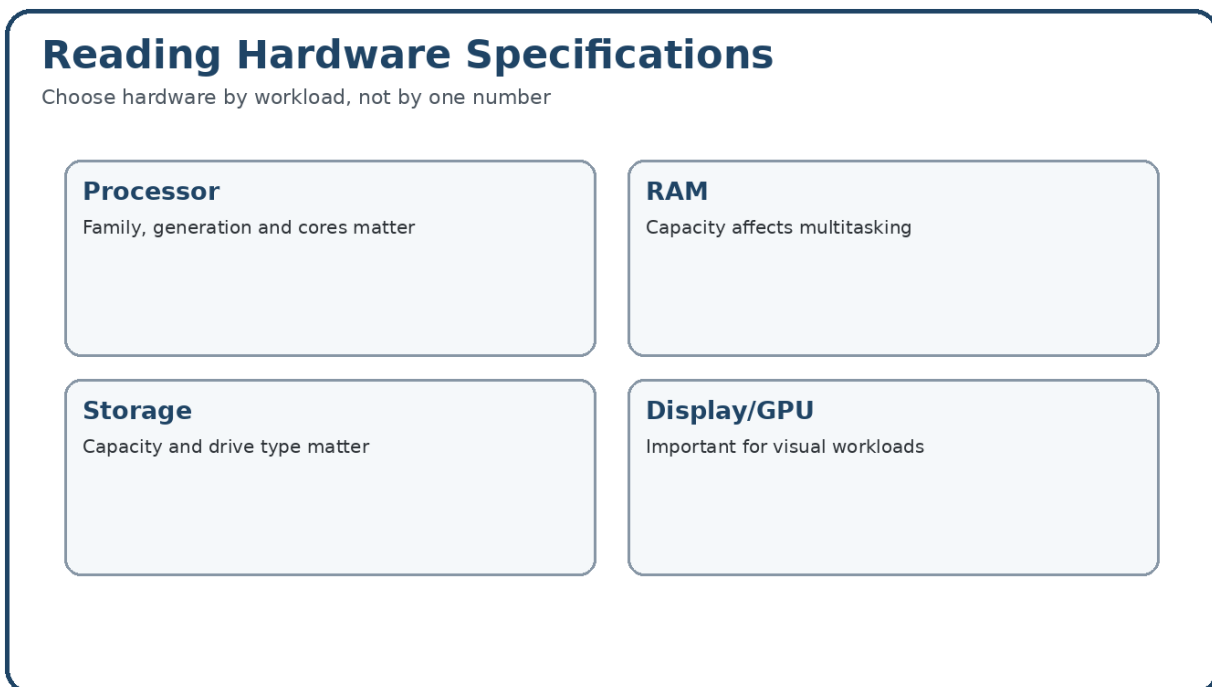


Figure 15: Instructional visual for Reading Computer Specifications.

Step-by-Step Guided Practical

1. Find the specification page for your own authorised computer in system settings.
2. Record processor, RAM and storage.
3. Create a fictional specification for an office learner.
4. Create a stronger fictional specification for video editing and explain the differences.

Ethan Digital Academy Case Study

The Academy can avoid overspending by matching computer specifications to each training room's real software requirements.

Review Questions

76. What is the main purpose of reading computer specifications?
77. Name two important hardware concepts from this lesson.
78. What safety or compatibility mistake should be avoided?
79. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 15 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 16: Hardware Compatibility and Upgrades

Introduction

An upgrade replaces or adds hardware to improve capacity or capability. Compatibility must be confirmed before purchasing or installing components.

Learning Objectives

- Explain the major concepts in hardware compatibility and upgrades.
- Complete the safe guided practical for hardware compatibility and upgrades.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: RAM upgrades require compatible memory type, capacity support and physical form factor.

Key point: Storage upgrades require a supported interface and physical space.

Key point: Power and cooling requirements matter for some expansion devices.

Key point: Manufacturer documentation is the safest reference for model-specific compatibility.

Key point: Learners should not dismantle devices unless the exercise is supervised on safe training equipment.

Step-by-Step Guided Practical

1. Use a fictional computer specification supplied by the instructor.
2. Identify which RAM type it supports from the worksheet.
3. Choose a compatible fictional storage upgrade from two options.
4. Explain what documentation should be checked before purchase.

Ethan Digital Academy Case Study

Before upgrading Academy computers, the technician should confirm model support rather than assuming every RAM module or drive fits every machine.

Review Questions

80. What is the main purpose of hardware compatibility and upgrades?
81. Name two important hardware concepts from this lesson.
82. What safety or compatibility mistake should be avoided?
83. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 16 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 17: Basic Hardware Troubleshooting

Introduction

Basic troubleshooting is a structured process of identifying symptoms, checking simple causes and escalating when specialist repair is required.

Learning Objectives

- Explain the major concepts in basic hardware troubleshooting.
- Complete the safe guided practical for basic hardware troubleshooting.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Describe the symptom precisely: no power, no display, no sound, no network, or slow performance.

Key point: Check safe external factors first, such as power, cables, brightness, volume and normal settings.

Key point: Restarting through the operating system can resolve some temporary problems.

Key point: Change one safe variable at a time so the cause is easier to identify.

Key point: Stop and escalate if there is burning smell, damaged power equipment, liquid exposure or a repair requiring internal electrical work.

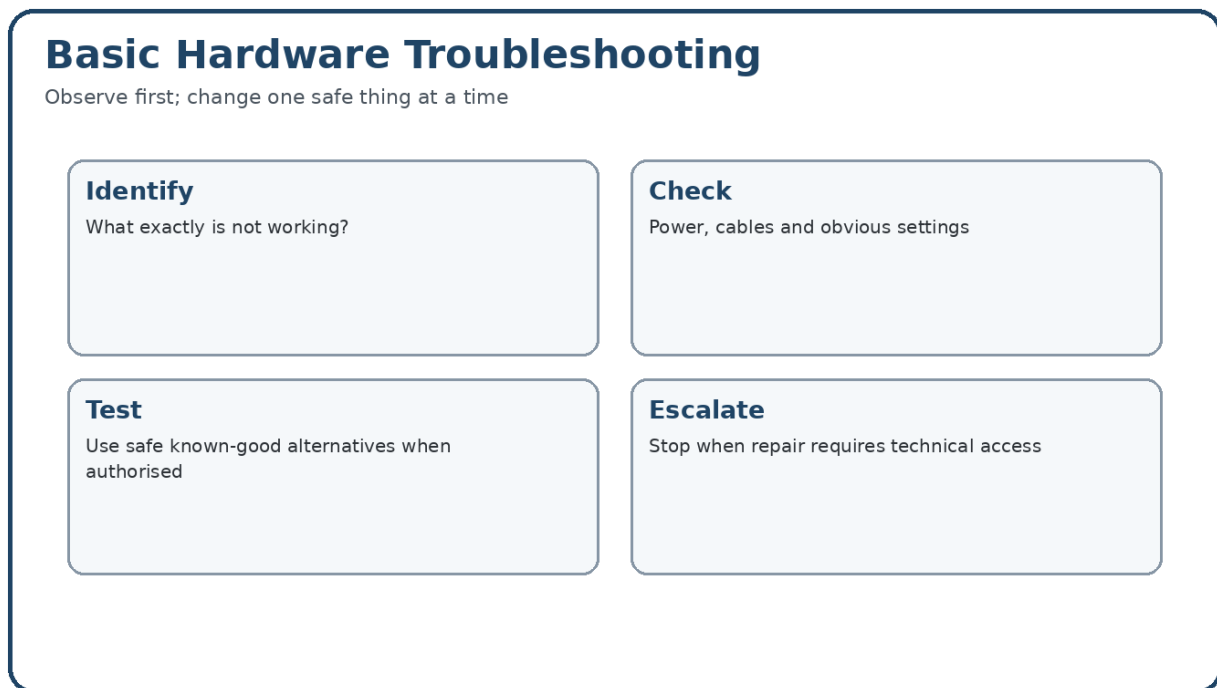


Figure 17: Instructional visual for Basic Hardware Troubleshooting.

Step-by-Step Guided Practical

1. Use a fictional 'monitor shows no picture' scenario.
2. List safe checks: power indicator, display cable, correct input and brightness.
3. Use a fictional 'no sound' scenario and list volume/output checks.
4. Write when you would stop and call a technician.

Ethan Digital Academy Case Study

Academy support staff can solve simple connection problems quickly while referring hardware faults to qualified technicians.

Review Questions

84. What is the main purpose of basic hardware troubleshooting?
85. Name two important hardware concepts from this lesson.
86. What safety or compatibility mistake should be avoided?
87. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 17 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 18: Preventive Care and Hardware Records

Introduction

Preventive care helps equipment remain reliable and makes maintenance easier to plan.

Learning Objectives

- Explain the major concepts in preventive care and hardware records.
- Complete the safe guided practical for preventive care and hardware records.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Keep liquids away from computer equipment.

Key point: Maintain clear ventilation and a clean workspace.

Key point: Use normal shutdown and approved power protection.

Key point: Record device model, serial/asset number and warranty information where authorised.

Key point: Back up important data because hardware can fail unexpectedly.

Step-by-Step Guided Practical

1. Create a fictional asset record with device name, model, RAM, storage and purchase date.
2. Inspect the workstation for blocked vents or loose external cables.
3. Organise cables safely without disconnecting critical equipment.
4. Write a monthly external-check checklist.

Ethan Digital Academy Case Study

An Academy asset register helps management know which computers are available, their specifications and maintenance history.

Review Questions

88. What is the main purpose of preventive care and hardware records?
89. Name two important hardware concepts from this lesson.
90. What safety or compatibility mistake should be avoided?
91. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 18 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Lesson 19: Final Computer Hardware Project

Introduction

The final project requires learners to recommend a computer system for a realistic Ethan Digital Academy training scenario.

Learning Objectives

- Explain the major concepts in final computer hardware project.
- Complete the safe guided practical for final computer hardware project.
- Relate hardware choices to real academic and workplace needs.

Teaching Content

Key point: Start with the user's workload rather than choosing the most expensive parts.

Key point: Balance processor, RAM, storage, graphics, display, ports and power protection.

Key point: Explain desktop versus laptop choice.

Key point: Include peripherals and backup considerations.

Key point: Use specifications, not unsafe hands-on dismantling, to justify the design.

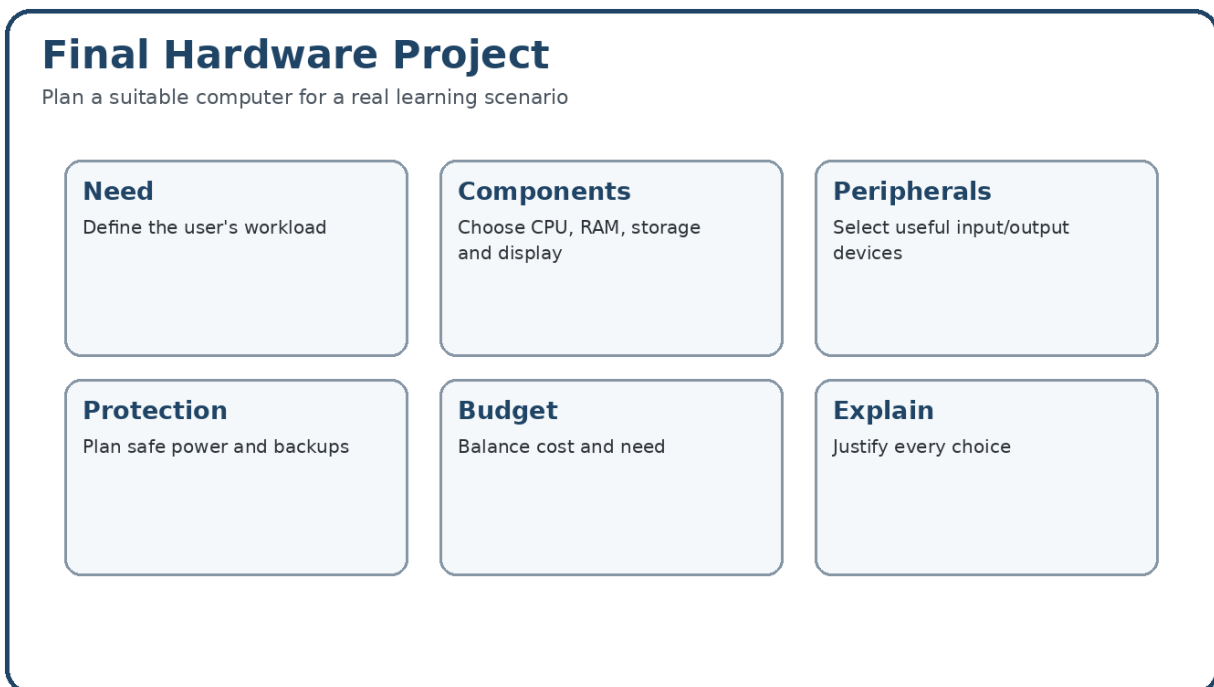


Figure 19: Instructional visual for Final Computer Hardware Project.

Step-by-Step Guided Practical

1. Choose one scenario: office training lab, digital-marketing workstation or beginner coding lab.
2. Write the required tasks.
3. Select a suitable CPU class, RAM amount, storage type/capacity and display.
4. Choose keyboard, mouse, networking and power protection.
5. Prepare a one-page justification and short presentation.

Ethan Digital Academy Case Study

This project demonstrates that hardware knowledge is about making appropriate, safe and cost-aware technology decisions.

Review Questions

92. What is the main purpose of final computer hardware project?
93. Name two important hardware concepts from this lesson.
94. What safety or compatibility mistake should be avoided?
95. How would this knowledge help when choosing or supporting a computer?

Assignment

Complete the Lesson 19 worksheet or practical using only safe, authorised equipment. Write 100-150 words explaining what you learned and one real-world use.

Final Practical Project - 100 Marks

Component	Marks	Requirement
Needs analysis	15	Define the intended workload and users.
Processor and RAM	20	Choose and justify appropriate capability.
Storage	15	Choose type and capacity.
Graphics/display	10	Match visual hardware to workload.
Ports/peripherals/network	15	Select useful connectivity and devices.
Power/protection	10	Include safe power and backup considerations.
Budget/value	10	Balance cost with need.
Presentation	5	Explain choices clearly.

Course Quiz - 30 Questions

1. What is computer hardware?
2. Name the four basic hardware functions.
3. What is a computer form factor?
4. What is the motherboard?
5. What is the purpose of the CPU?
6. What is a CPU core?
7. What is RAM?
8. Why is RAM called volatile memory?
9. What is an SSD?
10. State one difference between SSD and HDD.
11. Name three input devices.
12. Name three output devices.
13. What is USB used for?
14. What does HDMI commonly carry?
15. What is Ethernet used for?
16. What is a PSU?
17. What is a UPS used for?
18. Why does a computer need cooling?
19. What is integrated graphics?
20. What is a dedicated GPU?
21. What is display resolution?
22. What is a network interface?
23. What does a router do?
24. State one advantage of a desktop.
25. State one advantage of a laptop.
26. Why should specifications be matched to workload?
27. What must be checked before a hardware upgrade?

28. Name two safe first checks when a monitor shows no picture.
29. When should a hardware problem be escalated to a technician?
30. Why is an asset register useful?

Final Examination - 100 Marks

Section A - Core Components (20 marks)

Explain motherboard, CPU, RAM, storage and how they work together.

Section B - Peripherals and Connectivity (20 marks)

Classify input/output devices and explain common ports, cables and networking hardware.

Section C - Power, Cooling and Safety (20 marks)

Explain PSU, batteries, surge protection, UPS, cooling and safe hardware practice.

Section D - Specifications and Upgrades (20 marks)

Compare two fictional computers and recommend one for a stated workload, including compatibility considerations.

Section E - Troubleshooting and Planning (20 marks)

Describe a safe troubleshooting process and design a suitable computer for an Ethan Digital Academy learning scenario.

Course Completion Outcome

After successful completion, the learner should be able to identify major computer hardware, explain how core components work together, recognise ports and peripherals, interpret basic specifications, compare computer types, understand safe power and cooling practices, perform simple external troubleshooting checks and recommend appropriate hardware for common academic and workplace needs.

Professional Practice Checklist

- Never open powered equipment during beginner practicals.
- Never force connectors into ports.
- Match hardware to workload and compatibility requirements.
- Protect equipment with appropriate power practices.
- Keep ventilation clear.
- Back up important data.
- Escalate electrical or internal hardware faults to qualified personnel.